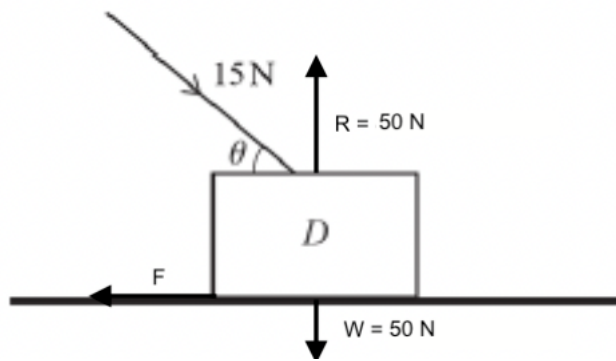


Forces 4

1.
 - i. $10 \cos 60^\circ = 5\text{N}$
 - ii. 1N
 - iii. $F = ma; 1 = 2a \therefore a = 0.5\text{ms}^{-2}$
 - iv. $S = ut + \frac{1}{2}at^2 = \frac{t^2}{4}$
 - v. $R = 2g - 10 \sin 60^\circ = 10.9$
2.
 - a.
 - i. $R = 98 + 10 \sin 20^\circ = 101.42\text{N}$
 - ii. $\mu = \frac{10 \cos 20^\circ}{101.42} = 0.0927$
 - b.
 - i. $R = 98 - 15 \sin 60^\circ = 85\text{N}$
 - ii. $\mu = \frac{15 \cos 60^\circ}{85} = 0.0882$
3.
 - i. $F = 9$
 - ii. $F = 0$
 - iii. $F = 9 \cos 60^\circ = 4.5\text{N}$
4.
 - a.
$$\begin{aligned}4 \sin 2\theta &= 6 \sin \theta \\8 \sin \theta \cos \theta &= 6 \sin \theta \\8 \cos \theta &= 6 \\ \theta &= \arccos \frac{3}{4} \\ &= 41.4^\circ\end{aligned}$$
 - b.
$$\begin{aligned}4 \cos 2\theta + 6 \cos \theta &= P \\4 \cos 82.8^\circ + 6 \cos 41.4^\circ &= P \\P &= 5\end{aligned}$$
 - c. Horizontal component $= 3 \sin 41.4^\circ = 1.98$
Vertical component $= -3 \cos 41.4^\circ + 5 = 2.75$
 - i. $\sqrt{(3 \sin 41.4^\circ)^2 + (-3 \cos 41.4^\circ + 5)^2} = 3.39\text{N}$
 - ii.
$$\begin{aligned}\tan \theta &= \frac{-3 \cos 41.4^\circ + 5}{3 \sin 41.4^\circ} \\ \theta &= 54.2^\circ\end{aligned}$$

5. a.



b. $\mu = 0.2$. Then For D to remain at rest, the horizontal force must be less than or equal to μR which is the maximum force that friction can exert on D .

$$15 \cos \theta \leq 0.2 \times (50 + 15 \sin \theta)$$

$$15 \cos \theta \leq 10 + 3 \sin \theta$$

$$15 \cos \theta - 3 \sin \theta \leq 10$$